

## **Project 5: Cross-Department Discharge Coordination with MCP, RBAC and PHI Boundaries**

### **Project Brief**

Build a cross-department discharge coordination system using three FastMCP servers:

EHR Server

Pharmacy Server

Billing Server

The system should coordinate patient discharge while enforcing tool-level RBAC and PHI boundaries. The discharge coordination agent should resolve brand-to-generic medication mismatches, check pharmacy stock, suggest alternatives, generate a validated invoice, and ensure clinical notes do not cross into billing context.

The core lesson is protocol-level governance: data boundaries should be enforced by tools and permissions, not merely by prompt instructions.

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### **Use-case Description with Examples and Health-tech Applicability**

Patient discharge often requires coordination between clinical, pharmacy, and billing teams. Manual handoffs can cause delays, errors, privacy risks, and billing disputes.

Example problem:

- EHR has discharge medication: “Tab Augmentin 625 mg”
- Pharmacy has generic inventory: “Amoxicillin + Clavulanate”
- Billing needs invoice code and amount
- Billing clerk should not see diagnosis or controlled substance details
- Pharmacy should not see unnecessary billing information

The system should coordinate safely:

1. Read discharge medication list from EHR
2. Convert brand to generic
3. Check pharmacy stock
4. Suggest alternatives if out of stock
5. Generate billing line items without exposing clinical notes
6. Validate invoice

## 7. Log all tool calls

HealthTech relevance:

- Healthcare interoperability
- MCP-based agent tooling
- PHI-safe workflow automation
- Discharge coordination
- Pharmacy and billing integration
- Healthcare data governance

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### Real-world Scenario with Data Snippets

#### EHR server data

```
{
  "patient_id": "P-1007",
  "discharge_status": "ready",
  "diagnosis_summary": "Community acquired pneumonia",
  "discharge_medications": [
    {
      "brand_name": "Augmentin",
      "dose": "625 mg",
      "frequency": "BD",
      "duration": "5 days"
    },
    {
      "brand_name": "Crocin",
      "dose": "500 mg",
      "frequency": "SOS",
      "duration": "3 days"
    }
  ]
}
```

```
],  
  "clinical_notes": "Patient improved after IV antibiotics..."  
}
```

#### **Pharmacy server data**

```
{  
  "inventory": [  
    {  
      "generic_name": "Amoxicillin + Clavulanate",  
      "strength": "625 mg",  
      "stock": 0,  
      "alternatives": ["Amoxicillin + Clavulanate 1g"]  
    },  
    {  
      "generic_name": "Paracetamol",  
      "strength": "500 mg",  
      "stock": 120  
    }  
  ]  
}
```

#### **Billing server expected input**

```
{  
  "patient_id": "P-1007",  
  "billable_items": [  
    {  
      "item": "Amoxicillin + Clavulanate 1g",  
      "quantity": 10  
    },  
    {
```

```
"item": "Paracetamol 500 mg",  
  "quantity": 6  
}  
]  
}
```

Billing should not receive:

diagnosis\_summary

clinical\_notes

controlled substance prescription details beyond billable-safe representation

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### Recommended Dataset Approach

Create synthetic discharge datasets:

| Dataset element            | Suggested count |
|----------------------------|-----------------|
| Discharge cases            | 100–200         |
| Medication master          | 100–300         |
| Brand-generic mappings     | 100–300         |
| Pharmacy inventory records | 100–300         |
| Billing item catalog       | 100–300         |
| RBAC policies              | 20–40           |
| Edge cases                 | 30–50           |

Include traps:

| Trap                   | Example                                   |
|------------------------|-------------------------------------------|
| Brand-generic mismatch | Augmentin vs amoxicillin-clavulanate      |
| Out of stock           | prescribed drug unavailable               |
| Similar drug names     | dopamine vs dobutamine                    |
| Controlled substances  | billing should not see full clinical note |

| Trap                    | Example                                      |
|-------------------------|----------------------------------------------|
| Wrong department access | billing tries to call EHR clinical-note tool |
| Missing dosage          | invoice cannot be validated                  |
| Allergy conflict        | alternative drug conflicts with allergy      |
| Duplicate billing       | same medicine billed twice                   |

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### Architecture Choices to Explore

| Server              | Tools                                                     |
|---------------------|-----------------------------------------------------------|
| EHR MCP Server      | get_discharge_summary, get_medication_list, get_allergies |
| Pharmacy MCP Server | check_stock, resolve_generic, suggest_alternative         |
| Billing MCP Server  | create_invoice, validate_invoice, get_billing_code        |

RBAC examples:

| Role              | Allowed tools                        |
|-------------------|--------------------------------------|
| Discharge agent   | limited EHR, pharmacy, billing tools |
| Pharmacy context  | medication list, allergies, stock    |
| Billing context   | billable items only                  |
| Billing clerk     | invoice data, not clinical notes     |
| Clinical reviewer | clinical summary, medication list    |

Architecture:

Discharge coordination agent

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EHR MCP server

retrieve discharge meds and allergy-safe summary

↓

Pharmacy MCP server

resolve generic, check stock, suggest alternatives

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Billing MCP server

generate invoice from billable-safe items

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Validation

check quantities, prices, duplicates, PHI leakage

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Discharge packet

medication availability + invoice + audit trace

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## **Progressive Levels of Implementation**

### **Level 1: Simple discharge workflow**

- Read discharge meds
- Check stock
- Generate invoice
- No MCP or RBAC

### **Level 2: Three MCP servers**

- Implement EHR, Pharmacy, Billing FastMCP servers
- Define tools for each server
- Agent calls tools to complete discharge

### **Level 3: Brand-generic resolution**

- Add medication normalization
- Resolve brand-to-generic
- Handle spelling and synonym issues

### **Level 4: RBAC and PHI boundaries**

- Enforce tool-level permissions
- Prevent clinical notes from reaching billing
- Mask sensitive data where needed

**Level 5: Advanced discharge coordination**

- Out-of-stock alternatives
- Allergy checks
- Invoice validation
- Audit trace
- Error handling and human review

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**Evaluations to Consider**

| Metric                                                                                               | What it measures                                     |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| Workflow completion rate                                                                             | Discharges completed successfully                    |
| Medication matching accuracy                                                                         | Brand-to-generic resolution correctness              |
| Stock-check accuracy                                                                                 | Correctly identifies available and unavailable drugs |
| Alternative suggestion safety                                                                        | Alternative is valid and allergy-safe                |
| Invoice accuracy                                                                                     | Correct billable items and quantities                |
| RBAC violation rate                                                                                  | Unauthorized tool/data access prevented              |
| PHI leakage rate                                                                                     | Billing context receives no clinical notes           |
| Latency                                                                                              | Discharge workflow duration                          |
| Audit completeness                                                                                   | Every tool call logged                               |
| Error recovery                                                                                       | System handles missing or conflicting data           |
| Demo corner cases:                                                                                   |                                                      |
| EHR says Augmentin; pharmacy only knows Amoxicillin + Clavulanate.                                   |                                                      |
| Prescribed medication is out of stock. System should suggest alternative and flag for approval.      |                                                      |
| Billing agent asks for clinical notes. Tool-level RBAC should deny access.                           |                                                      |
| Patient has penicillin allergy. Alternative amoxicillin should be blocked.                           |                                                      |
| Same medication appears twice under brand and generic names. Invoice should avoid duplicate billing. |                                                      |

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## **What Participants Should Build and Deliver**

Participants should deliver:

1. Three FastMCP servers: EHR, Pharmacy, Billing
2. Tool definitions for each server
3. Tool-level RBAC rules
4. Discharge coordination agent
5. Brand-to-generic medication resolver
6. Pharmacy stock checker
7. Alternative suggestion logic
8. Invoice generator
9. Invoice validator
10. PHI boundary enforcement
11. Audit log of all tool calls
12. Demo showing successful discharge and blocked unauthorized access
13. Presentation deck